

RICHARD HUMULOCK

Senior Software Engineer · Distributed Systems & Platform Engineering
Baltimore, MD · rphumulock@gmail.com · richardhumulock.fly.dev · [LinkedIn](#) · [GitHub](#)
Active Top Secret Clearance

SUMMARY

Software engineer with 10+ years building scalable, data-driven systems end to end — distributed backends, event-driven architecture, and cloud infrastructure. Depth in Go and JVM services, messaging and streaming, and PostgreSQL performance, with a record of setting security and data-governance standards in cleared environments.

EXPERIENCE

Technical Marketing Engineer | Synadia · Baltimore, MD (Remote) [START DATE] – Present

- Design and build reference architectures and production-grade demo applications on NATS.io and Synadia Cloud, demonstrating event-driven patterns including pub/sub, JetStream persistence, and key-value and object stores.
- Author developer documentation, technical guides, and tutorials for adopting NATS in production.
- Work across the stack in Go, TypeScript, and hypermedia frameworks to demonstrate low-latency application patterns from broker to browser.

Senior Research Engineer | Two Six Technologies · Baltimore, MD (Remote) September 2020 – [END DATE]

- Architected scalable backend services with Vert.x and HyperExpress, deploying a Hazelcast-backed Vert.x cluster for resilience, redundancy, and distributed state management.
- Implemented secure authentication workflows using OAuth2 and PKCE, and designed and enforced Role-Based Access Control for granular, auditable access to protected resources.
- Established data governance practices in compliance with ISM standards, defining the controls the team built against.
- Optimized PostgreSQL schemas and query performance; integrated WebSockets into Ember.js for real-time updates.
- Mentored junior engineers and raised the team baseline on security and scalability practices.

Software Development Engineer | U.S. Engineering Solutions · Baltimore, MD March 2018 – September 2020

- Built real-time weather monitoring dashboards in React and Redux, reducing data load times by 30%.
- Developed Spring Boot and Hibernate services backed by Redis to cache and serve high-frequency weather data.
- Designed an AWS ingestion pipeline using Lambda for real-time processing and S3 for durable, scalable storage.

Earlier Eutality Group (2017–2018) · Mannakee Group (2016–2017) · e-Management (2014–2016) — C#/.NET physics simulation, AngularJS and SharePoint applications for high-security environments, and a Java web crawler with D3.

SELECTED PROJECTS

Distributed Sensor Cluster · Vert.x · Hazelcast · Datastar · Docker Compose

Three-node Vert.x cluster where virtual sensors are deployed and torn down at runtime. Nodes coordinate over the clustered event bus with Hazelcast managing membership; readings stream to the UI over SSE rather than polling.

Real-Time Multiplayer over Redis Pub/Sub · Node.js · WebSockets · Redis

Implemented the WebSocket handshake by hand rather than using a library, broadcasting moves through Redis Pub/Sub; game state persists so matches survive a server restart.

ECS Game Server · Go · entity-component-system

ECS design separating data from behavior; systems iterate contiguous component arrays for cache locality.

TECHNICAL SKILLS

Languages Go, TypeScript, JavaScript, Java, C#, SQL

Messaging NATS.io / JetStream, RabbitMQ, Redis Streams, Hazelcast, WebSockets, gRPC

Architecture Event-driven systems, microservices, event sourcing, CQRS, SAGA, serverless

Backend Vert.x, Spring Boot, Node.js (Express, Fastify), ASP.NET Core, REST, GraphQL

Data PostgreSQL, Redis, MongoDB, MySQL, schema design, query and index optimization

Frontend React, Vue.js, Svelte, Angular, Ember.js, Datastar, TailwindCSS

Cloud & DevOps AWS (Lambda, S3), Docker, Kubernetes / K3s, Fly.io, GitHub Actions, GitLab CI

Security OAuth2 / PKCE, JWT, Role-Based Access Control, ISM compliance

Testing Playwright, k6, Jest, JUnit

EDUCATION

B.S., Computer Science | University of Maryland, College Park, MD

2010 – 2014

A.S., Computer Science | College of Southern Maryland, La Plata, MD

2008 – 2010